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Bachelor's in Mathematics, option: Data Science

Project Title

Parkinson's Disease Prediction using MRI Images and Machine Learning

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List of abbreviations

AAL: Automated Anatomical Labeling

AUC: Area Under the Curve

BIDS: Brain Imaging Data Structure

BOLD: Blood Oxygen Level Dependent

CNN: Convolutional Neural Network

CSF: Cerebrospinal Fluid

DWT: Discrete Wavelet Transform

fMRI: Functional Magnetic Resonance Imaging

FN: False Negative

FP: False Positive

GA: Genetic Algorithm

GM: Gray Matter

HC: Healthy Control

KNN: K-Nearest Neighbors

MRI: Magnetic Resonance Imaging

MNI: Montreal Neurological Institute

PCA: Principal Component Analysis

PD: Parkinson’s Disease

PHOG: Pyramid Histogram of Oriented Gradients

ROI: Region of Interest

SBR: Striatal Binding Ratio

SMOTE: Synthetic Minority Over-sampling Technique

SPECT: Single Photon Emission Computed Tomography

SVM: Support Vector Machine

TN: True Negative

TP: True Positive

WM: White Matter

Abstract

Parkinson's disease is a progressive neurological disorder that affects movement and coordination due to the loss of dopamine-producing brain cells. Early diagnosis is important because it allows earlier treatment and better disease management. However, diagnosing Parkinson's disease remains difficult and usually depends on clinical examinations and medical expertise.

This project investigates the use of resting-state functional Magnetic Resonance Imaging (fMRI) and machine learning techniques for Parkinson's disease prediction. The dataset used in this study was obtained from the OpenNeuro platform and contains MRI data from 55 subjects, including healthy controls (HC) and Parkinson's disease (PD) patients. A preprocessing pipeline using Docker and fMRIPrep was applied to improve image quality and prepare the data for analysis. Functional connectivity matrices were computed from extracted Region of Interest (ROI) time series using Pearson correlation coefficients.

The connectivity matrices were transformed into numerical feature vectors and used as input for machine learning models. Ten classification models were evaluated, including SVM, Logistic Regression, Ridge Classifier, KNN, Random Forest, Gradient Boosting, XGBoost, LightGBM, Extra Trees, and SVM with RBF kernel. Feature selection, PCA dimensionality reduction, SMOTE balancing, and threshold optimization were applied to all models.

The results showed that linear models achieved the best overall performance. SVM Linear obtained the highest single-split test F1-score (0.875), while Logistic Regression achieved the highest AUC (0.893). A Soft Voting ensemble combining SVM Linear, Logistic Regression, and Ridge Classifier achieved the best clinical performance, with Recall = 0.825 during repeated cross-validation and Recall = 1.00 on the held-out test set.

Overall, the results demonstrate that resting-state MRI data can be effectively used for Parkinson's disease prediction using machine learning techniques, showing promising reliability for supporting early detection.

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1 Introduction

Parkinson disease is a progressive disorder of the nervous system which affects movement due to the loss of dopamine producing brain cells. And worsens over time, affecting the brain's ability to control movement and coordination. The symptoms can differ from one person to another. Early symptoms may be mild, and you may not even notice them. Symptoms often begin on one side of the body, then affect both sides, they can also be similar to symptoms of other disorders.

Early detection of this disease is crucial, as it allows for better management of symptoms, and allows patients to adopt preventative measures earlier. However, diagnosing Parkinson disease remains a challenging task, as it primarily relies on efficient medical tests and neurologists with high expertise.

Magnetic Resonance Imaging (MRI) is one of the commonly used medical imaging techniques in the evaluation of neurological disorders. It is a medical test that uses magnets and radio waves to produce detailed images of the brain and surrounding tissues, helping diagnose neurological disorders, injuries, and structural abnormalities. It is relatively accessible and non-invasive, making it an attractive option for supporting diagnosis. However, despite its usefulness, MRI alone is not considered a reliable technique for medical diagnosing, as its accuracy remains moderate and varies depending on many factors.

With the advancement of artificial intelligence, machine learning techniques have shown strong potential in medical analysis. These techniques can automatically extract useful patterns and insights from MRI data that may not be easily detectable by human experts. This opens the possibility of improving the accuracy and the diagnostic value of MRI by transforming it from a regular medical tool into a predictive system.

The main objective of this project is to investigate whether MRI images can be effectively used to predict Parkinson's disease using machine learning techniques and evaluating its reliability. A web application will be implemented that allows the user to submit his MRI brain image and get a prediction of the probability of Parkinson's disease.

2 Literature Review

2.1 KNN (K- Nearest Neighbor)

K-Nearest Neighbors (KNN) is a machine learning algorithm used for many classification tasks including Parkinson's disease prediction. It decides the class of data points based on the closest similar points.

Approach 1: In this case a modified K-Nearest Neighbors (KNN) algorithm for Parkinson's disease detection is proposed which involves multimodal data, including gait |(movement), handwriting, and voice features. The model improves **traditional** KNN by incorporating a weighting mechanism instead of simple voting process, which makes it more sensitive to data distribution. The proposed approach achieved high classification accuracy, reaching 99.60% for gait data, 97.8% for handwriting, and 94.5% for voice data. The results show that the modified KNN gives improved performance over traditional classifiers and it is effective across different dataset sizes (1).

Approach 2: Another research paper used the K-Nearest Neighbors (KNN) algorithm for Parkinson's disease detection using MRI brain images. The model uses the PHOG descriptor to extract the manually designed features and applied **multiple** feature selection methods before classification. **KNN** was used alongside **SVM** in a **Hybrid** classification model, and the system was evaluated using 10-fold cross-validation. The final predictions were obtained using an Iterative Majority Voting (IMV) strategy to combine outputs from different classifier–feature combinations. This approach achieved high performance, with overall accuracy exceeding 98.5%, reflecting a strong performance in the Parkinson's disease classification (2).

The accuracy of the 2 approaches is compared, showing a higher one for the combined KNN and SVM model (Table 2.1).

Table 2-1 KNN approaches comparison.

Approach	Data	Methods	Accuracy
Modified KNN	Gait, voice, handwriting	Weighted KNN	99.6%, 97.8%, 94.5%
MRI-Based KNN	Brain MRI images	MRI-based (KNN + SVM) hybrid machine learning model	>98.5%

2.2 Logistic Regression

Logistic Regression is a supervised machine learning algorithm used for binary classification, often being applied to medical issues including diagnosing diseases through probability estimation from a given set of features.

Approach 1: The study proposes a machine learning approach based on biomedical data for Parkinson's disease prediction. This begins with data preprocessing, including filtering, normalization, and noise removal, to improve data quality. Then, feature selection using the LAFS algorithm is performed to select key symptoms such as tremors, speech problems, and movement disorders.

Classification is done using the Logistic Decision Regression model which differentiates healthy from unhealthy subjects. The model is compared with SVM and CNN regarding specificity, sensitivity, and performance at different data sizes (Figure 2-1). It performs well with a 93% sensitivity, 95% specificity, and ~96% classification performance, outperforming SVM (~55–60%) and CNN (~48–50%) (3).

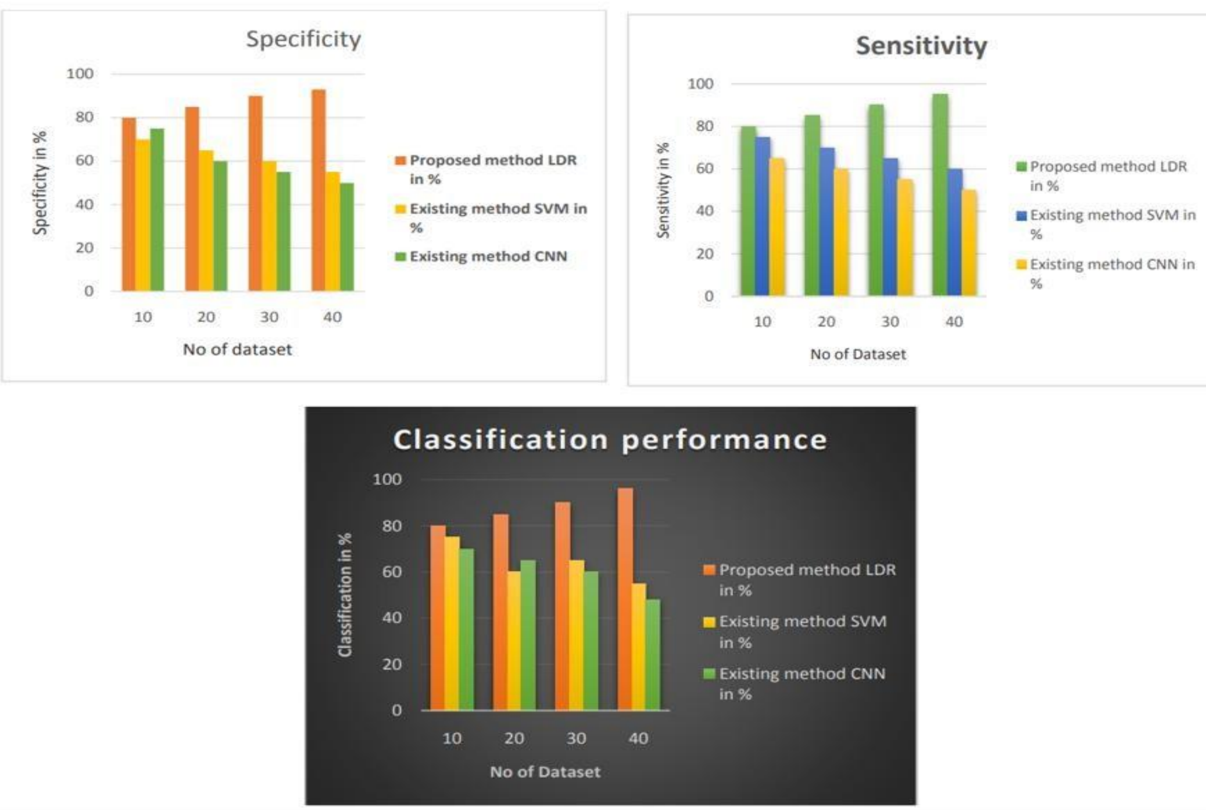


Figure 2-1 Sensitivity, Specificity, and Performance Comparison of LDR, SVM, and CNN

Approach 2: The study suggests a machine learning technique for the early detection of Parkinson's disease using logistic regression based on SPECT imaging data.

This technique relies on four SBR features (caudate and putamen of left and right brain) (Figure 2-2), which reduces complexity while keeping the most relevant information.

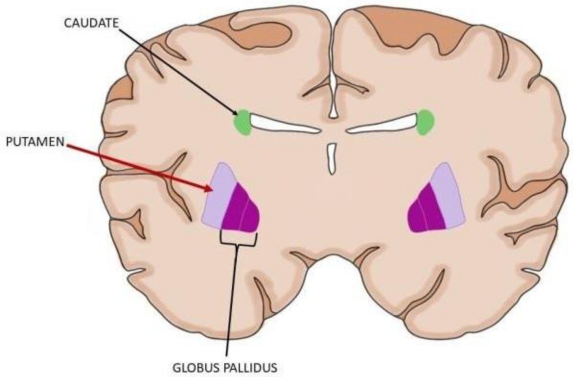


Figure 2-2 Caudate and Putamen of the brain

A logistic regression model is then built to estimate the probability of Parkinson's disease, enabling classification and risk prediction.

The model is trained on the large and diverse PPMI dataset, improving robustness, and is tested against SVM, showing strong and statistically significant performance for early diagnosis support (4).

2.3 Support Vector Machine (SVM)

Approach 1: The study applies a Support Vector Machine (SVM) to classify Parkinson's disease patients using voice measurement data from the UCI dataset.

First, data preprocessing technique through Weka is used, where correlated and irrelevant features are removed to improve data quality. Then, the SVM classifier is trained with various kernel functions (linear, polynomial, RBF, sigmoid) to find the best classification performance.

The model is evaluated using **cross-validation**, analyzing metrics such as true positive rate, false positive rate, and ROC curves. Results reveal that the linear kernel achieves the best performance (~65% accuracy), and performance improves with higher cross-validation folds (5).

Approach 2: This study proposes an integrated approach combining **Genetic Algorithm (GA)** and **Support Vector Machine (SVM)** for Parkinson's disease detection. using **speech signal data**, specifically voice recordings of patients pronouncing the vowel sound "a".

First, the speech signals are transformed using **Discrete Wavelet Transform (DWT).** Then, multiple mathematical features are extracted, then a Genetic Algorithm (GA) is implemented to select the most relevant features and optimize the model.

Finally, an SVM classifier is used for classification, achieving an accuracy of 91% (6). A comparison table is performed, to compare the 2 approaches on different aspects (Table 2.2).

Table 2-2 Two SVM approaches models' comparison

Aspect	Approach 1	Approach 2
Data	Voice dataset (UCI)	Speech signals ("a")
Model	SVM (Kernels)	Integrated (GA + SVM)
Accuracy	~65%	~91%
Complexity	Low	High

3 Methodology

3.1 Dataset Description

The dataset used in this project was obtained from **OpenNeuro** platform, a publicly available source for neuroimaging data. The dataset contains 55 different subjects. It can be accessed through the following link: https://openneuro.org/datasets/ds005892/versions/1.0.0/download

The graph below presents the class distribution of the dataset, showing the balance between Parkinson's disease cases and control subjects (Figure 3-1).

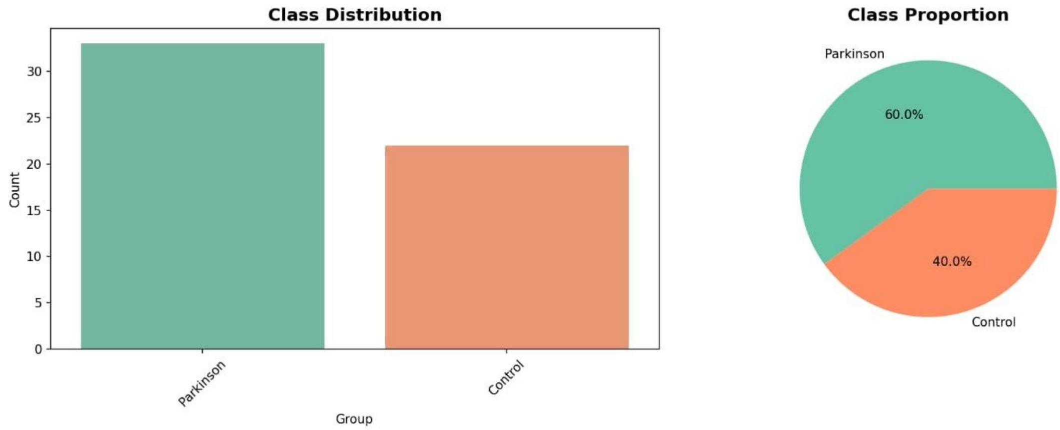


Figure 3-1 Class Distribution of the dataset

This dataset follows the **Brain Imaging Data Structure** (BIDS) standard, which organizes neuroimaging data in a consistent and structured format. Each subject (patient) in the dataset contains multiple folders representing different types of brain imaging data.

3.2 Dataset Structure

For each subject, the dataset includes the following main components:

- **anat**
  This folder contains anatomical MRI scans (T1-weighted images). These images mainly provide high-resolution structural information about the brain and are used for brain structure analysis, spatial normalization, and alignment with functional data (Figure 3-2).

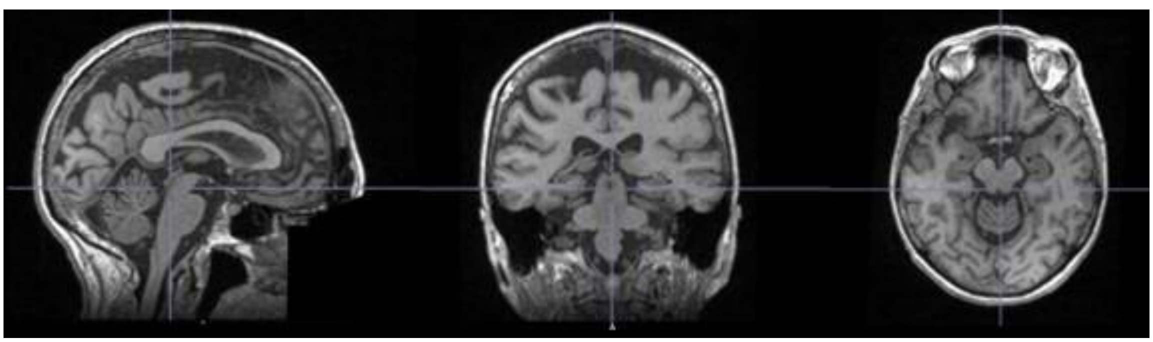


Figure 3-2 Anatomical brain image

- **func**
  This folder contains functional MRI (fMRI) data (BOLD signals). These scans capture brain activity over time by measuring changes in blood oxygen levels. They are mainly used for analyzing brain activity patterns and supporting predictive modeling (Figure 3-3).

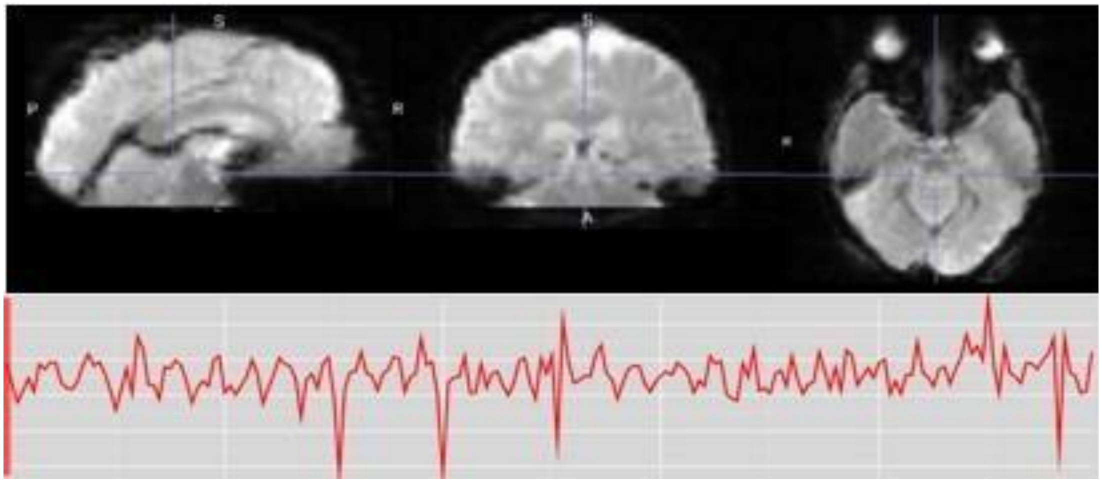


Figure 3-3 Functional brain image with time series graph

- **fmap**
  This folder contains field maps used for correcting distortions in MRI images. These are important for improving image quality and correcting spatial distortions in functional MRI data.

➢ **Additional elements that are included in the dataset:**

- **Dataset_description.json:**
  Dataset-level metadata file providing general information about the dataset, including its **name, version (BIDS 1.8.0), authors, license, study context, funding, references, DOI, and keywords**.
- **participants.json:**
  Metadata file describing the variables in participants.tsv, including field definitions and categorical levels (e.g., HC = Healthy Control, PD-NC, PD-MCI, M/F).
- **participants.tsv:**
  Tab-separated file containing participant-level data, including participant ID, group (HC / PD-NC / PD-MCI), age, and sex.
- **README:**
  File providing general information about the dataset, including usage notes and important context.
- **CHANGES:**
  File documenting updates and modifications made to the dataset over time.

This structured organization allows efficient preprocessing and integration of both anatomical and functional data.

3.3 Preprocessing Objective and Tools

To prepare the dataset for analysis, a preprocessing pipeline was applied using **Docker** and the **fMRIPrep** tool in it. The goal of this preprocessing step is not only to improve image quality but also to transform raw **MRI** data into **structured** and **meaningful** representations that can be used as input for machine learning models. Instead of working directly with raw images, the preprocessing pipeline extracts relevant features and reduces noise, resulting in data that is more suitable for model training and prediction.

3.4 Preprocessing Environment

Docker is a platform that allows applications to run in a controlled and reproducible container, ensuring consistent results across different systems. Within this environment, **fMRIPrep** was used as an automated preprocessing tool specifically designed for neuroimaging data, providing a robust and standardized pipeline for processing both anatomical and functional **MRI** data.

3.5 Anatomical Processing Pipeline (T1-weighted Images)

- **Intensity Non-Uniformity Correction**
  MRI images often suffer from intensity variations caused by scanner imperfections. This step corrects these variations to produce a uniform image.
- **Skull Stripping (Brain Extraction)**
  non-brain tissues such as the skull and scalp were removed to isolate the brain region so that only the brain is used in further preprocessing phases. This step is essential for accurate analysis and reduces noise in the data (Figure 3-4).

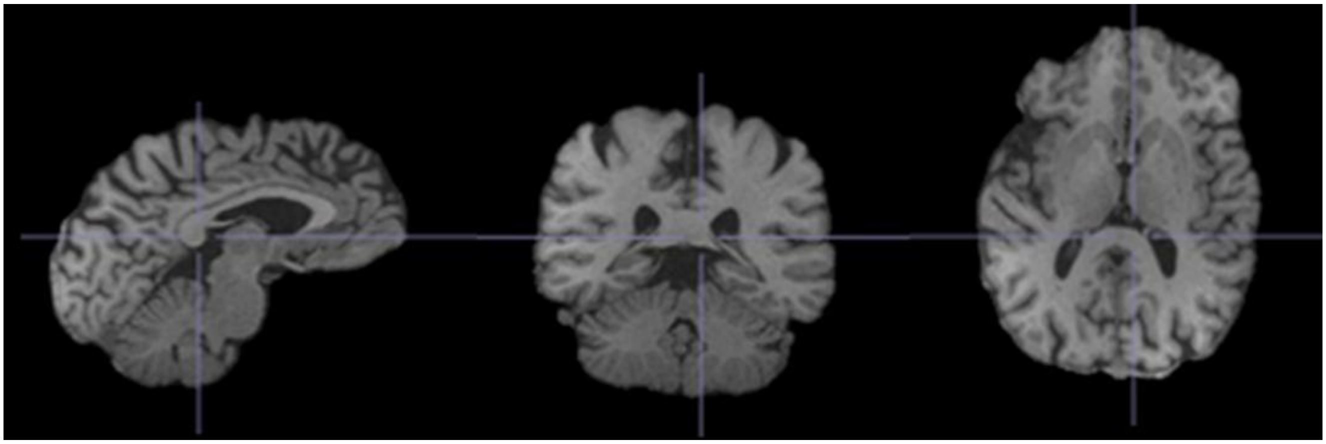


Figure 3-4 Anatomical brain extracted image

- **Tissue Segmentation**
  The brain was segmented into three main tissue types: Gray Matter (GM), White Matter (WM), and Cerebrospinal Fluid (CSF). This helps in understanding brain structure and supports further processing steps.

- **Spatial Normalization to Standard Space (MNI)**
  Each brain image was aligned to a standard brain template (MNI space). This allows comparison across subjects and ensures consistent spatial representation, so that all brain images are aligned to the same template (MNI152), meaning the same location in different brains corresponds to the same brain region (Figures 3-5 and 3-6).

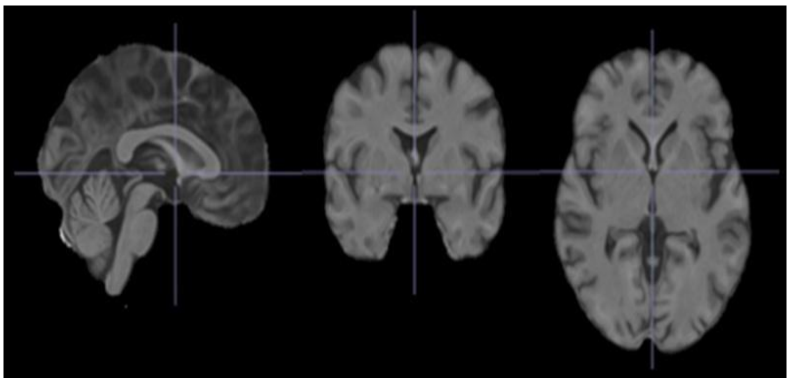


Figure 3-5 Anatomical brain image aligned to MNI152 template

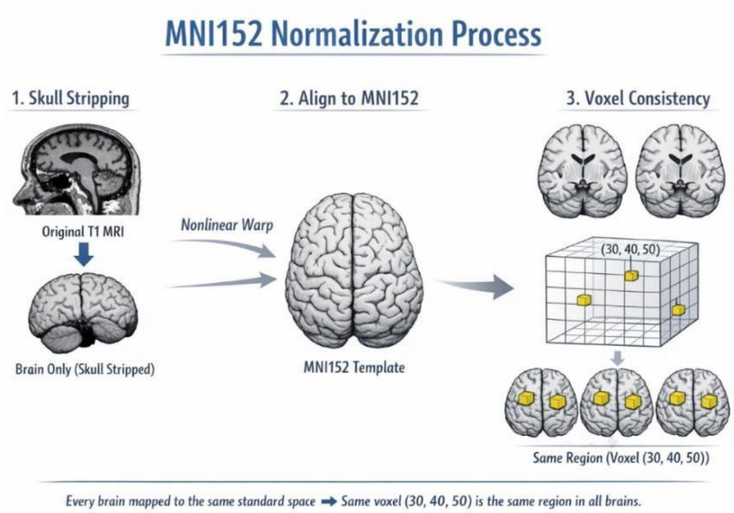


Figure 3-6 MNI152 normalization process

3.6 Functional Processing Pipeline (BOLD fMRI)

The functional MRI data passed through several preprocessing steps to ensure accuracy and reliability:

- **Generation of Reference Volume**
  A reference image was generated from the functional data to serve as a stable baseline for subsequent processing steps.

- **Head Motion Correction**
  Since patients may move during scanning, motion correction was applied to align all volumes and reduce movement-related distortions (Figure 3-7).

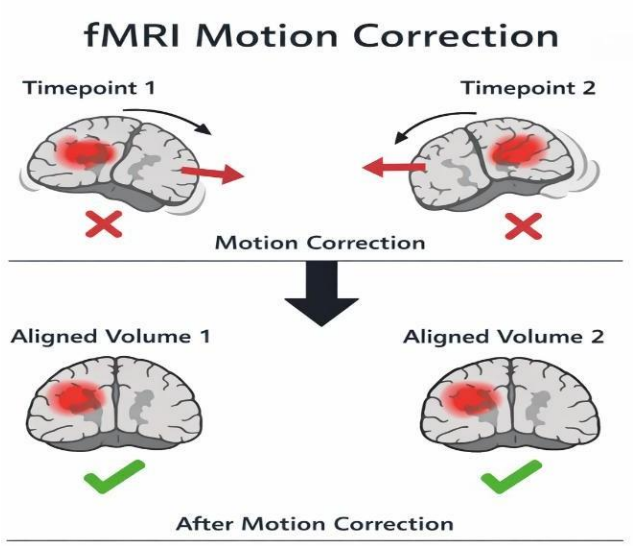


Figure 3-7 fMRI Motion correction

- **Slice Timing Correction**
  This step corrects timing differences between slices in each scan, improving temporal consistency.
- **Susceptibility Distortion Correction**
  Distortions caused by magnetic field inhomogeneities were corrected using the available field map data, improving spatial accuracy.
- **Co-registration with Anatomical Images**
  Functional images were aligned with anatomical images to ensure accurate spatial correspondence between structure and function (Figure 3-8).

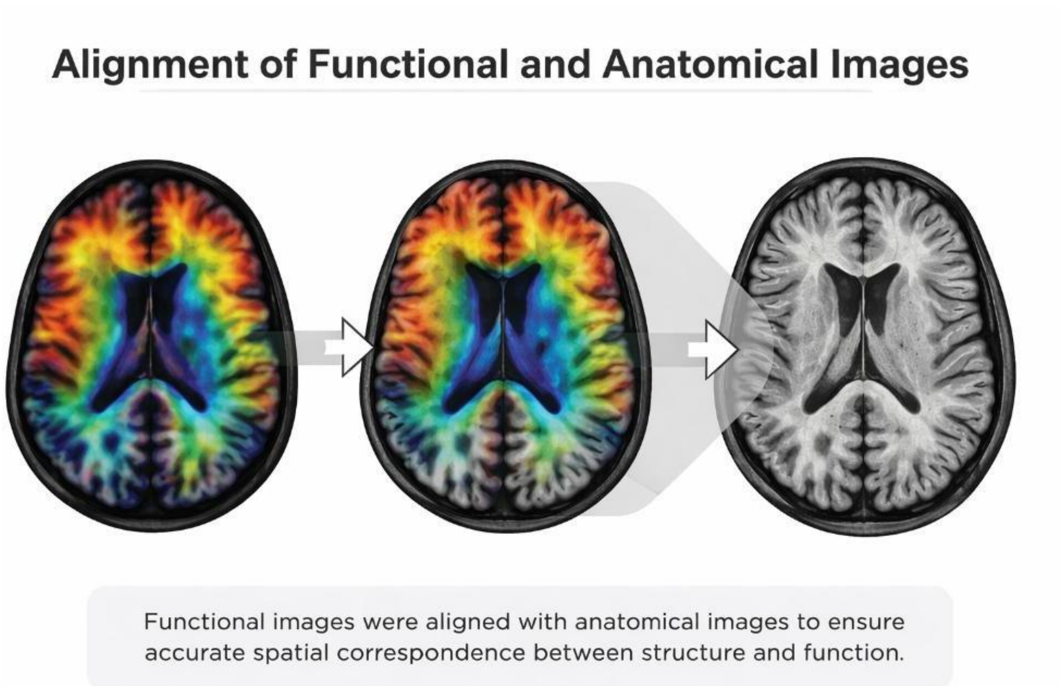


Figure 3-8 Functional and Anatomical images alignment

- **MNI152 Normalization**
  Functional data was transformed into the standard MNI152 space used for anatomical data, ensuring consistency across subjects.

- **Brain Mask Creation**
  A brain mask was generated to isolate brain regions and remove non-brain areas from the analysis (Figure 3-9).

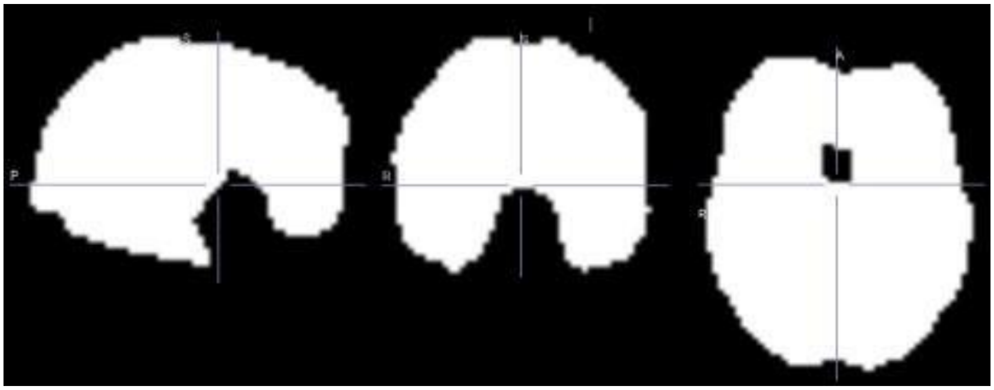


Figure 3-9 Functional image brain mask

- **Confound Extraction**
  Noise-related variables were extracted, including head motion parameters and physiological signals from white matter and cerebrospinal fluid. These confounds help reduce noise and improve the quality of the data used for modeling.

3.7 Data Loading and Input Preparation

The loaded input data will be the preprocessed fMRI data of size 4D in MNI152 space and an anatomical brain atlas. The fMRI data is 4D, where the first three dimensions represent the brain space and the fourth dimension represent time.

The **AAL** atlas is a **3D label image**, where each voxel contains a regional index corresponding to a brain area (Figure 3-10).

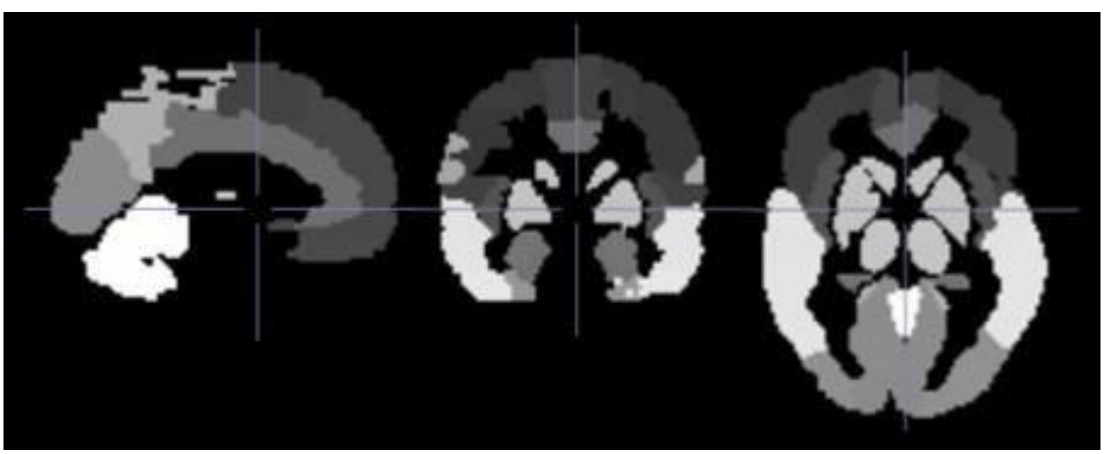


Figure 3-10 AAL_116 Atlas

3.8 Atlas Resampling

As the fMRI data and the atlas are defined in different spatial resolutions and may differ in several properties (Figure 3-11), resampling of the atlas is required to make sure that both images are aligned to the same coordinate space.

Property	fMRI	Atlas
Shape	(50, 62, 48, 200)	(91, 109, 91)
Resolution	3.125 × 3.125 × 3.5 mm	2 × 2 × 2 mm
Data Type	4D (spatial + time)	3D (region labels)
Orientation	Standard	Flipped (X-axis)
Origin	(-77.25, -113.25, -78)	(90, -126, -72)
Grid Size	Coarser	Finer

Figure 3-11 Key Differences Between fMRI and Atlas

This step uses:

- a preprocessed fMRI image in MNI space
- an anatomical atlas containing labelled brain regions

For example:

- fMRI shape: (50, 62, 48, 200))
- Atlas shape: (91, 109, 91)

This ensures that each atlas label correctly corresponds to the appropriate brain voxels in the fMRI image (Figure 3-12).

Nearest-neighbor interpolation is applied when resampling the atlas since the atlas contains integer labels representing regions in the brain. Unlike linear interpolation, this method preserves original label values and doesn't corrupt the anatomical region labels.

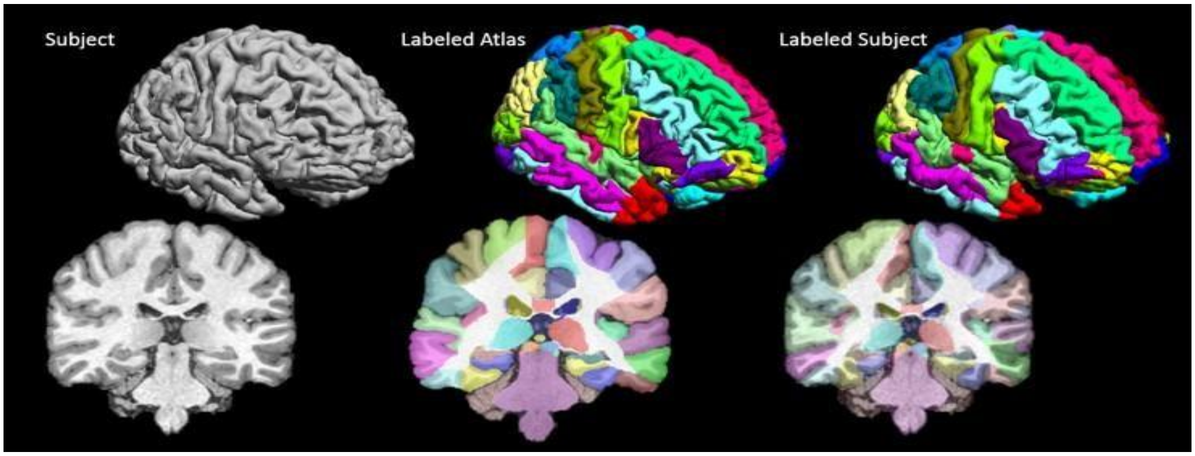


Figure 3-12 Subject brain, Original Atlas, and Aligned Atlas

3.9 ROI Time Series Extraction

- Once alignment is completed, Region of Interest (ROI) time series will be derived from the fMRI data. Each atlas label represents a specific brain region, and all voxels belonging to a brain region are chosen.

- For each region, the average value of BOLD signal is obtained for each brain region across all voxels at each time point, giving one representative time series per brain region.

- The resulting output is a matrix of shape (T, R), where T is the number of time points and R is the number of brain regions (Figures 3-13 and 3-14).

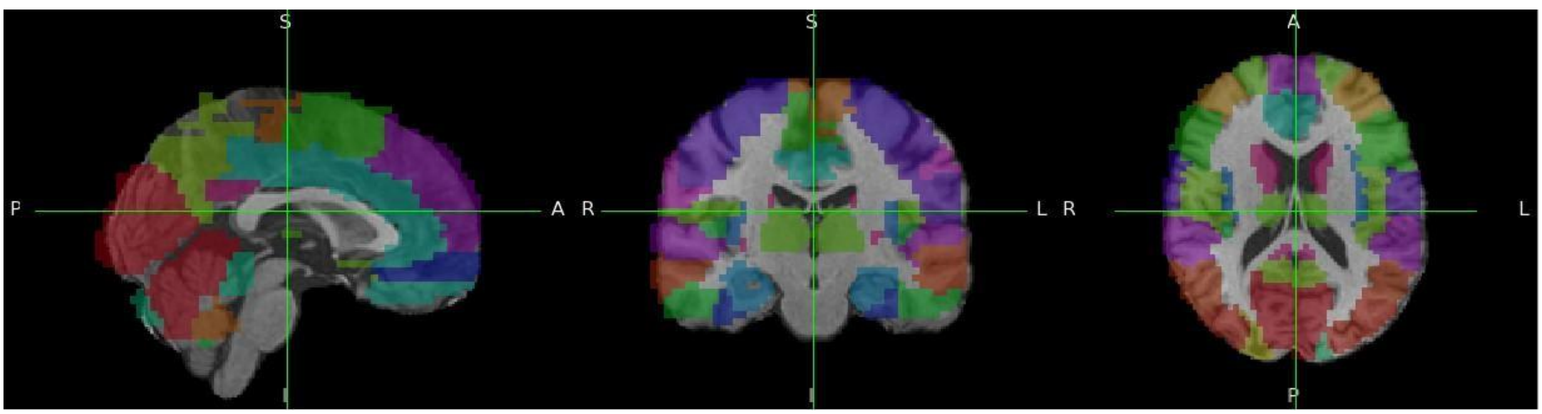


Figure 3-13 Brain Atlas Mapped onto Anatomical Image

	ROI_2001	ROI_2002	ROI_2101	ROI_2102	ROI_2111	ROI_2112	ROI_2201	ROI_2202	ROI_2211	ROI_2212	...	ROI_9081	ROI_9082	ROI_9100
0	679.297794	736.598931	727.767477	766.002206	408.632699	412.058224	730.967542	744.595030	572.996922	567.877303	...	103.693277	264.010298	559.882013
1	672.745041	728.048035	721.014302	755.940982	405.765934	409.849507	726.119146	735.766321	571.960190	563.337320	...	106.143712	262.809949	547.773010
2	668.541325	723.332179	718.986515	752.234513	405.038882	407.673010	722.692504	730.976264	569.704806	563.113989	...	97.058090	255.876216	550.564342
3	669.387732	723.173497	718.354604	750.220907	402.910561	406.317822	721.629122	729.435495	567.549005	560.634990	...	100.124165	260.832547	541.817288
4	669.307330	722.871353	718.683245	750.962168	404.057227	405.630631	723.202734	729.989650	566.964227	559.057618	...	100.532042	259.675383	555.483693

5 rows × 116 columns

Figure 3-14 ROI values

3.10 Functional Connectivity Matrix Computation

- Using the extracted ROI time series, functional connectivity is computed to measure the relationships between brain regions. This is done using the **Pearson** correlation coefficient to find how similar the time series of any two ROIs are.

Pearson correlation coefficient: $$r = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}}$$

➢ Elements of the formula

- r: Pearson correlation coefficient
- x_i: value of variable **X** for observation *i*
- y_i: value of variable **Y** for observation *i*
- $\bar{x}$: mean (average) of all **X values**
- $\bar{y}$: mean (average) of all **Y values**
- $(x_i - \bar{x})$: deviation of each X value from its mean
- $(y_i - \bar{y})$: deviation of each Y value from its mean
- $\sum$: summation over all observations (add all values)
- **Numerator:** $\sum(x_i - \bar{x})(y_i - \bar{y})$
  measures how X and Y vary **together** (covariance-like)
- **Denominator:** $\sqrt{\sum(x_i - \bar{x})^2 \sum(y_i - \bar{y})^2}$
  normalizes by variability (standard deviations)

➢ Short interpretation

- $r \in [-1,1]$
- **+1** → perfect positive relationship
- **0** → no linear relationship
- **−1** → perfect negative relationship

- This outcome is a symmetric connectivity matrix of shape (R, R), where each element represents the correlation between two brain regions. The diagonal values are equal to 1, and all values range between -1 and 1.

- This matrix represents the functional interactions within the brain, and it is the basis for upcoming feature extraction.

And here is an example of a Functional connectivity heatmap for a patient (Figure 3-15):

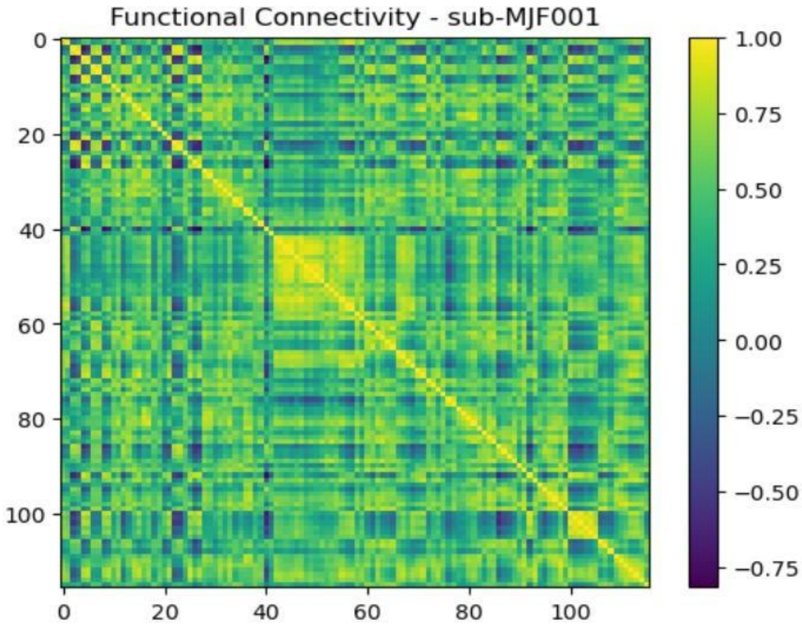


Figure 3-15 Functional Connectivity Heatmap for a patient

3.11 Feature Extraction and Dataset Preparation

After computing the functional connectivity matrix for each subject, the next step is to extract numerical features that can be used as input for machine learning models.

- As the functional connectivity matrix is symmetric, only the upper triangular part (excluding the diagonal) is considered to eliminate redundant information.
- These values are flattened into a one-dimensional feature vector for each subject. As a result, each subject is represented by a collection of features based on their unique brain region connectivity.
- The final dataset has a shape of (N, F), where N is the number of subjects and F is the number of extracted connectivity features (Figure 3-16). For this study, the resulting dataset consists of 55 subjects with 6670 features per subject, which will be used as input for classification machine learning models.

#	Subject	0	1	2	...	6666	6667	6668	6669
1	sub-MJF001	0.77516027581285	0.76940542821147	0.80341236135494	...	-0.1546138524745	0.94429221608194	0.82118899148693	0.90113583977863
2	sub-MJF002	0.85647129485420	0.65130357970187	0.72298990182862	...	0.89941206783496	0.98043040376890	0.91447041343197	0.88473717812128
3	sub-MJF003	0.96763790213010	0.60410161764324	0.67627775799144	...	-0.6926816163439	-0.5498502794574	-0.5197430608321	0.98010497992703
				- - - 49 rows hidden - - -					
53	sub-MJF079	0.90187379007815	0.78004502972749	0.79614701156948	...	0.74376078803005	0.96608370158281	0.80706958616845	0.83527260403188
54	sub-MJF080	0.84443266776248	0.61285930472729	0.68699972385916	...	0.19951839332982	0.91010224702442	0.03683022975561	-0.0035374349164
55	sub-MJF082	0.85421375601481	-0.1463579055232	0.07747154589701	...	0.45621845361199	0.83626723358321	0.39299761501224	0.57501925979655

Figure 3-16 Extracted Feature Matrix

4 Modelling and Results

This section presents the classification results obtained from applying 10 machine learning models to resting-state fMRI functional connectivity data for Parkinson's disease detection.

Each model was independently optimized through a structured pipeline incorporating feature selection, dimensionality reduction, class balancing via SMOTE, and per-model threshold optimization.

Performance is evaluated on a test set (80/20 stratified split) and reported alongside repeated stratified **cross-validation** estimates (5-fold × 10 repeats = 50 folds) for **stability**. An additional soft voting ensemble of the top 3 models is evaluated and compared against individual classifiers.

4.1 Experimental Setup

The significant class imbalance (PD: HC ≈ 1.9:1) and high dimensionality relative to sample size necessitated a preprocessing pipeline applied identically to every model (Figure 4-1):

- **Variance Threshold** — removed zero- or near-zero-variance features before scaling.
- **RobustScaler** — normalized features using median and interquartile range, reducing sensitivity to fMRI motion artefacts.
- **SMOTE** — applied within each training fold only to balance the HC/PD ratio, preventing synthetic samples from contaminating validation folds.
- **SelectKBest** (mutual_info_classif) — retained the top K most informative connectivity features; K tuned per model via grid search.
- **PCA** — applied after feature selection for linear models, retaining 85–95% of explained variance.
- **Threshold optimization** — classification threshold swept from 0.25 to 0.75, selecting the value maximizing F1-score on CV folds.

Hyperparameters were selected via **GridSearchCV** with 5-fold stratified cross-validation scored by F1. All tree-based models (RandomForest, Extra Trees, Gradient Boosting, XGBoost, LightGBM) were trained without PCA.

The primary evaluation metric is the **F1-score**, appropriate for imbalanced clinical data. Secondary metrics include AUC, precision, recall, accuracy and the confusion matrix.

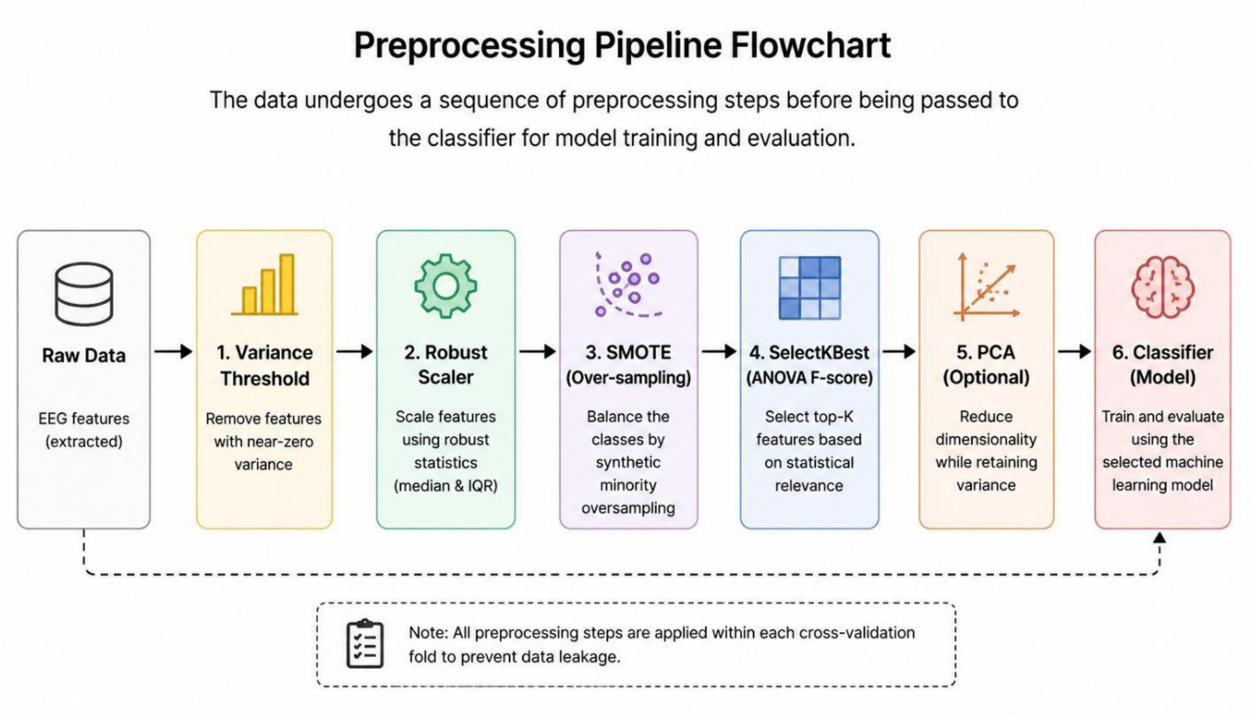


Figure 4-1 Models Preprocessing Pipeline Flowchart

4.2 Overall Models Performance

This section shows the single-split test set performance of the 10 models, sorted by **F1-score** (Table 4-1)**.**

Models are evaluated on the held-out test set of 11 subjects (4 HC, 7 PD). Green shading indicates models with TN ≥ 2 (genuine HC–PD discrimination). Yellow shading indicates models where TN < 2 (biased toward predicting PD).

Table 4-1 Test set performance of all models

Model	F1	AUC	Precision	Recall	Accuracy	Threshold
#1 SVM Linear	0.8750	0.8571	0.778	1.0000	0.8182	0.40
#2 Logistic Regression	0.8571	0.893	0.8571	0.8571	0.8182	0.40
#3 Ridge Classifier	0.8571	0.8571	0.8571	0.8571	0.8182	0.44
#4 Gradient Boosting	0.824	0.643	0.7000	1.0000	0.7273	0.50
#5 SVM RBF	0.824	0.8571	0.7000	1.0000	0.7273	0.27
#6 Random Forest	0.824	0.7143	0.7000	1.0000	0.7273	0.27
#7 KNN	0.8000	0.804	0.7500	0.8571	0.7273	0.35
#8 Extra Trees	0.824	0.786	0.7000	1.0000	0.7273	0.39
#9 XGBoost	0.7500	0.786	0.667	0.8571	0.6364	0.45
#10 LightGBM	0.7500	0.786	0.667	0.8571	0.6364	0.41

Interpretation: The findings indicate that Linear classifiers (SVM Linear, Logistic Regression, Ridge Classifier) had the highest F1 scores (0.857–0.875) with strong AUC values (0.857–0.893), implying that the fMRI functional connectivity features are linearly separable in the reduced feature space. Distance classifiers (KNN) and ensemble methods (RandomForest) obtained moderate F1 scores (0.800). Boosting-based models (Gradient Boosting, SVM RBF) achieved similar F1 (0.824) but with TN=1, this implies that they were biased towards the majority class.

The gradient boosting family (XGBoost, LightGBM) performance was the worst (F1=0.750), consistent with the known limitation of boosting methods on small datasets.

The figure below compares the statistical metrics of the 10 models (Figure 4-2):

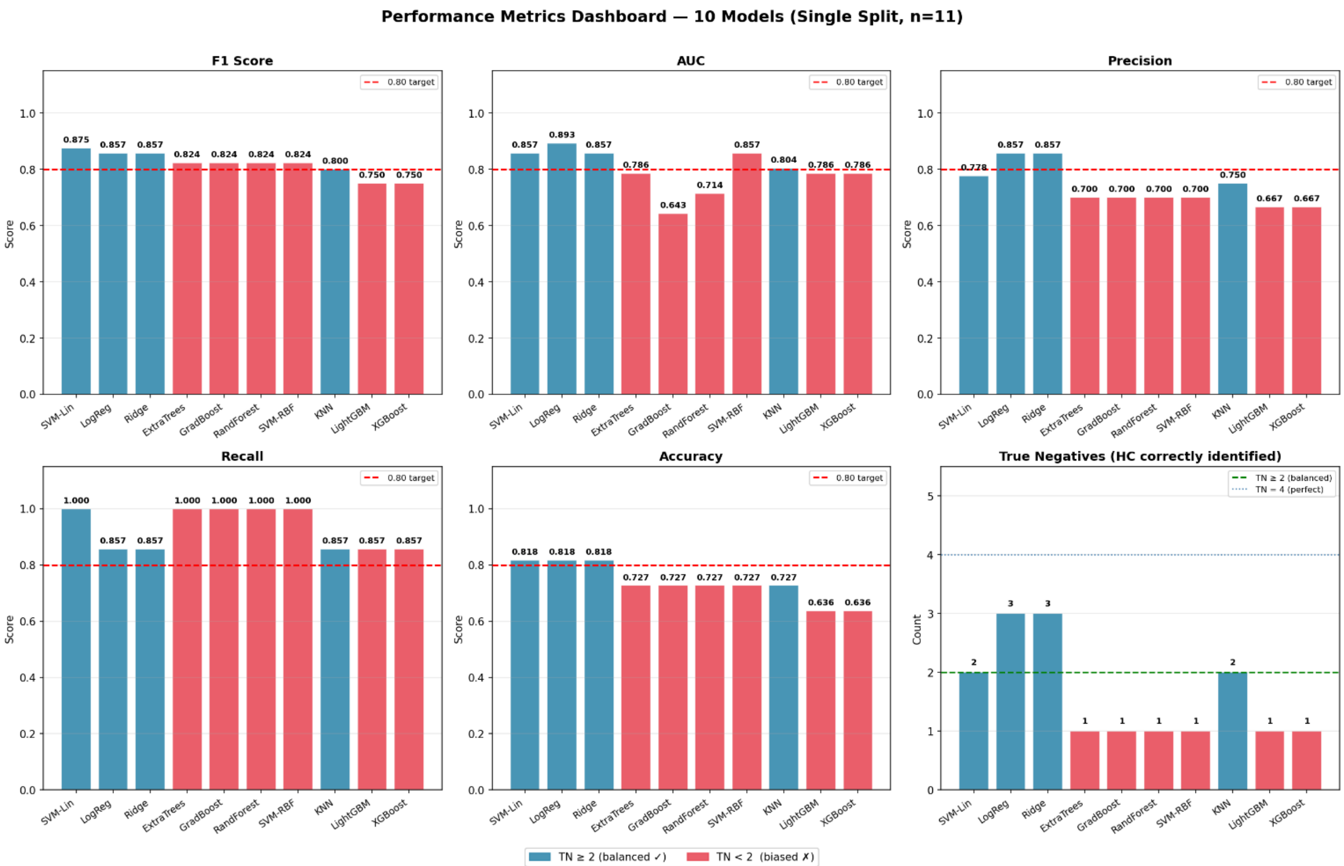


Figure 4-2 Metrics Dashboard for 10 models

4.3 Models Analysis

➢ SVM Linear

<u>**Configuration:**</u> The best-performing hyperparameters for the SVM Linear model are:

- **Kernel:** Linear
- **C:** 0.1
- **Class weight:** balanced
- **SelectKBest k:** 40

- **PCA variance:** 85%
- **Threshold:** 0.40

Test Set Performance:

The best results in terms of F1 on test set were achieved by the SVM Linear classifier with F1=0.875 and AUC=0.857. It is important to note that the model produced **zero false negatives (FN=0)**, meaning that all seven cases of PD were correctly classified. This makes it the most clinically valuable model in terms of sensitivity, as missing a PD diagnosis (false negative) carries greater clinical risk than a false positive. Two healthy subjects were misclassified as PD (FP=2), and two were correctly identified (TN=2).

Confusion matrix (Figure 4-3):

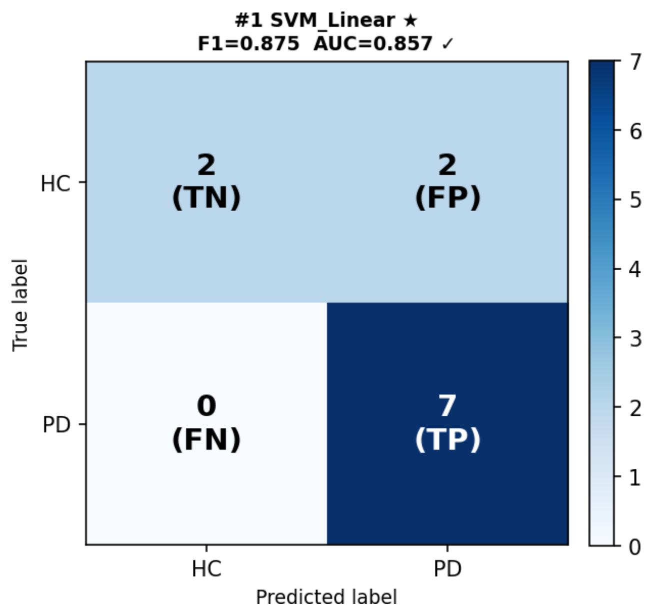


Figure 4-3 SVM_Linear Confusion Matrix

Classification report (Figure 4-4):

```
#1  SVM_Linear
Test F1=0.8750  AUC=0.8571  Threshold=0.40
Confusion: TN=2 FP=2 FN=0 TP=7

              precision    recall  f1-score   support

          HC       1.00      0.50      0.67         4
          PD       0.78      1.00      0.88         7

    accuracy                           0.82        11
   macro avg       0.89      0.75      0.77        11
weighted avg       0.86      0.82      0.80        11
```

Figure 4-4 SVM_Linear Classification Report

Interpretation:

The use of a linear kernel is aligned with the high-dimensionality characteristic of functional connectivity data, where linear separability is well-established in neuroimaging literature. The selection of 40 features by SelectKBest and 85% PCA variance retention implies that a wide range of connectivity played a role in the model's classification process, instead of only relying on a certain group of features. The balanced class weight ensured equal penalization of HC and PD misclassifications during training.

➢ Logistic Regression

Configuration: The best-performing hyperparameters for the Logistic Regression model are:

- **Penalty:** ElasticNet (L1_ratio=0.7)
- **C:** 1
- **Class weight:** balanced
- **SelectKBest k:** 25
- **PCA variance:** 85%
- **Threshold:** 0.40

Test Set Performance:

Logistic Regression showed the highest F1 score of 0.857 and the highest AUC score of 0.893 out of all individual models. Strangely, it showed the most balanced classification report per-class, with F1=0.75 for HC and F1=0.86 for PD. Confusion matrix shows (TN=3, FP=1, FN=1, TP=6) which is a good example of bidirectional learning, since it classified correctly 3 out of 4 HC subjects and 6 out of 7 PD subjects. The ElasticNet penalty (l1_ratio=0.7) imposed sparsity, effectively performing additional feature selection within the model.

Confusion matrix (Figure 4-5):

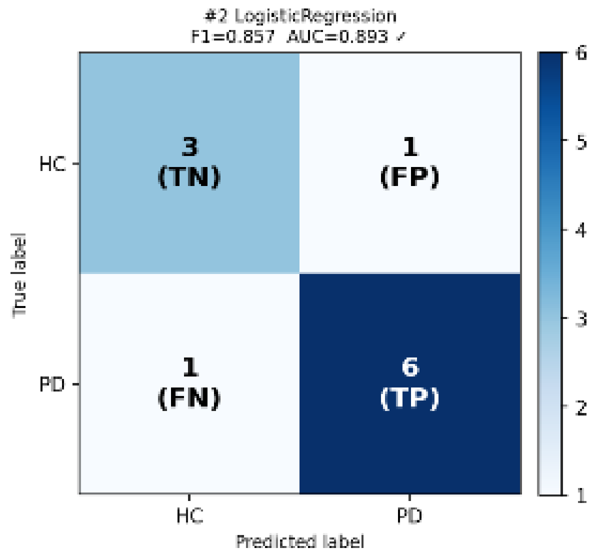
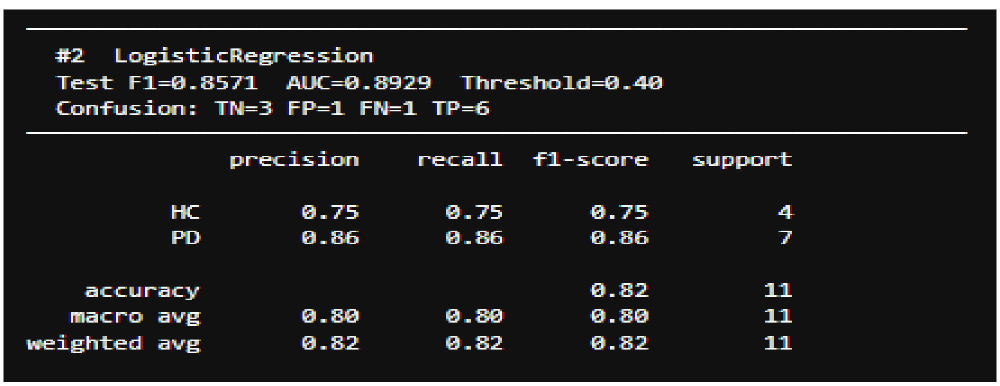


Figure 4-5 Logistic Regression Confusion Matrix

Classification report (Figure 4-6):



```
#2  LogisticRegression
Test F1=0.8571  AUC=0.8929  Threshold=0.40
Confusion: TN=3 FP=1 FN=1 TP=6

              precision    recall  f1-score   support

          HC       0.75      0.75      0.75         4
          PD       0.86      0.86      0.86         7

    accuracy                           0.82        11
   macro avg       0.80      0.80      0.80        11
weighted avg       0.82      0.82      0.82        11
```

Figure 4-6 Logistic Regression Classification Report

<u>Interpretation:</u>

Based on the high AUC, the probabilities generated by the logistic regression classifier are well-calibrated, which is particularly valuable when the model is used as a component in the soft voting ensemble. The macro-average F1 of 0.80 is the highest achieved by all models, showing a balanced performance across both classes.

➢ Ridge Classifier

Configuration: The best-performing hyperparameters for the Ridge Classifier model are:

- **Alpha:** 50.0
- **Class weight:** HC:3, PD:1}
- **SelectKBest k:** 25
- **PCA variance:** 85%
- **Threshold:** 0.44

Test Set Performance:

The Ridge Classifier (wrapped in CalibratedClassifierCV for probability prediction) it gave F1-score=0.857 and AUC=0.857, producing results similar to Logistic Regression on the test set. The strong regularization parameter (α=50) **prevented overfitting**, considering the small number of samples used in the study. The class weight {HC:3, PD:1} assigned three times the penalty to HC misclassification, minimizing false negatives (where it was TN=1 before).

Confusion matrix (Figure 4-7):

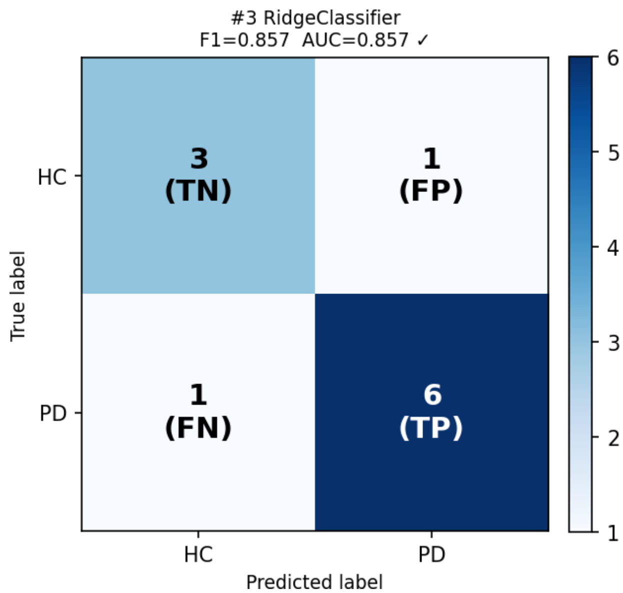


Figure 4-7 Ridge Classifier Confusion Matrix

Classification report (Figure 4-8):

```
#3  RidgeClassifier
Test F1=0.8571  AUC=0.8571  Threshold=0.44
Confusion: TN=3 FP=1 FN=1 TP=6

              precision    recall  f1-score   support

          HC       0.75      0.75      0.75         4
          PD       0.86      0.86      0.86         7

    accuracy                           0.82        11
   macro avg       0.80      0.80      0.80        11
weighted avg       0.82      0.82      0.82        11
```

Figure 4-8 Ridge Classifier Classification Report

Interpretation:

Ridge Classifier's strong performance proves that the issue of classification using fMRI connectivity is easily solved by a regularized linear classifier. The identical test-set performance to Logistic Regression shows that the two models are generating similar decision boundaries in the reduced feature space.

➢ Gradient Boosting

Configuration The best-performing hyperparameters for the Gradient Boosting model are:

- **n_estimators:** 100
- **max_depth:** 2
- **learning_rate:** 0.1
- **subsample:** 0.8
- **min_samples_leaf:** 4
- **SelectKBest k:** 40
- **Threshold:** 0.50

Test Set Performance:

Gradient Boosting algorithm gave F1=0.824 with an excellent recall value (Recall=1.000, FN=0), however the low AUC of 0.643 suggests that there is a problem with probability calibration. The confusion matrix (TN=1, FP=3) shows that only 1 of 4 HC subjects was correctly classified, which reflects the major drawback of this model. The macro F1 of 0.61 and HC F1 of 0.40 shows that the model is poor at HC classification

Confusion matrix (Figure 4-9):

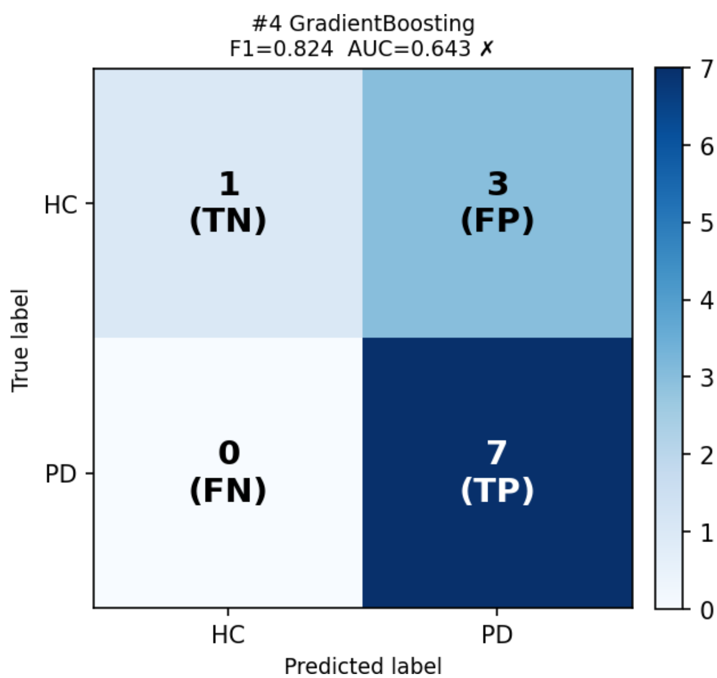


Figure 4-9 Gradient Boosting Confusion Matrix

Classification report (Figure 4-10):

```
 #4  GradientBoosting
 Test F1=0.8235  AUC=0.6429  Threshold=0.50
 Confusion: TN=1 FP=3 FN=0 TP=7

              precision    recall  f1-score   support

          HC       1.00      0.25      0.40         4
          PD       0.70      1.00      0.82         7

    accuracy                           0.73        11
   macro avg       0.85      0.62      0.61        11
weighted avg       0.81      0.73      0.67        11
```

Figure 4-10 Gradient Boosting Classification Report

Interpretation:

Despite significant improvements after running again with larger hyperparameter grids and class weights tuning, the Gradient Boosting model still somehow favoring PD predictions. This proves the known behavior of gradient boosting on small imbalanced datasets even after SMOTE balancing.

➢ SVM RBF

Configuration: The best performing hyperparameters for the SVM RBF model are:

- **Kernel:** RBF
- **C:** 5
- **gamma:** 0.01
- **Class weight:** balanced
- **SelectKBest k:** 15
- **PCA variance:** 85%
- **Threshold:** 0.27

Test Set Performance:

SVM with an RBF kernel achieved F1=0.824 and AUC=0.857, but like Gradient Boosting, didn't classify correctly most HC subjects (TN=1, FP=3). The low value of optimal threshold (0.27) means that the classifier provides low probability for even the most probable PD samples, which makes it necessary to have a low cutoff to obtain acceptable recall. Importantly, the post-fix approach showed degraded performance compared to the original one (AUC dropped from 0.857 to 0.179 in one configuration), indicating the sensitivity of the RBF kernel to SMOTE-augmented subspace and regularization techniques.

Confusion matrix (Figure 4-11):

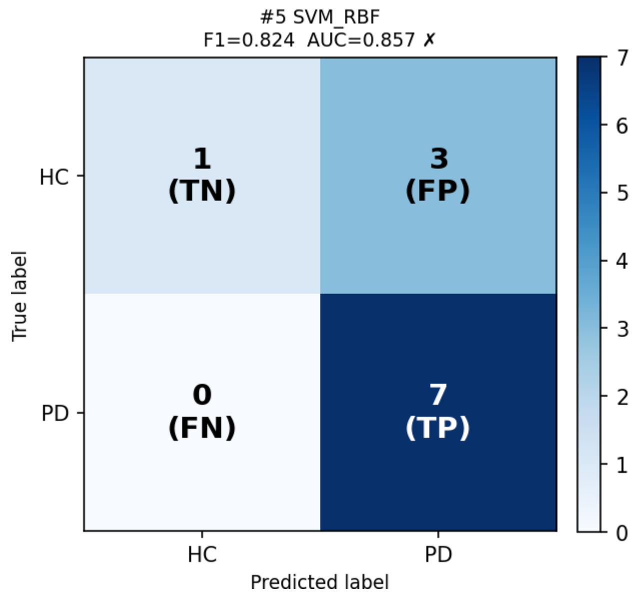


Figure 4-11 SVM_RBF Confusion Matrix

Classification report (Figure 4-12):

```
#5  SVM_RBF
Test F1=0.8235  AUC=0.8571  Threshold=0.27
Confusion: TN=1 FP=3 FN=0 TP=7

              precision    recall  f1-score   support

          HC       1.00      0.25      0.40         4
          PD       0.70      1.00      0.82         7

    accuracy                           0.73        11
   macro avg       0.85      0.62      0.61        11
weighted avg       0.81      0.73      0.67        11
```

Figure 4-12 SVM_RBF Classification Report

<u>Interpretation:</u>

Decision boundaries introduced by the RBF kernel are nonlinear, and thus the risk of overfitting will increase when applied on this small dataset. Linear kernel always showed better performance than the RBF kernel in terms of F1 and AUC, suggesting that changes related to PD are linearly separable.

➢ Random Forest

Configuration: The best-performing hyperparameters for the Random Forest model are:

- **n_estimators:** 100
- **max_depth:** 3
- **min_samples_leaf:** 4
- **Class weight:** balanced
- **SelectKBest k:** 40
- **Threshold:** 0.27

Test Set Performance:

Random Forest obtained F1-score=0.800 with a balanced confusion matrix (TN=2, FP=2, FN=1, TP=6). Although its F1 test equivalent to that of KNN, Random Forest shows a lower AUC (0.714) indicating weaker probability calibration. The shallow trees (max_depth=3) and high min_samples_leaf (4) show strong regularization required to prevent overfitting on 44 training subjects.

Confusion matrix (Figure 4-13):

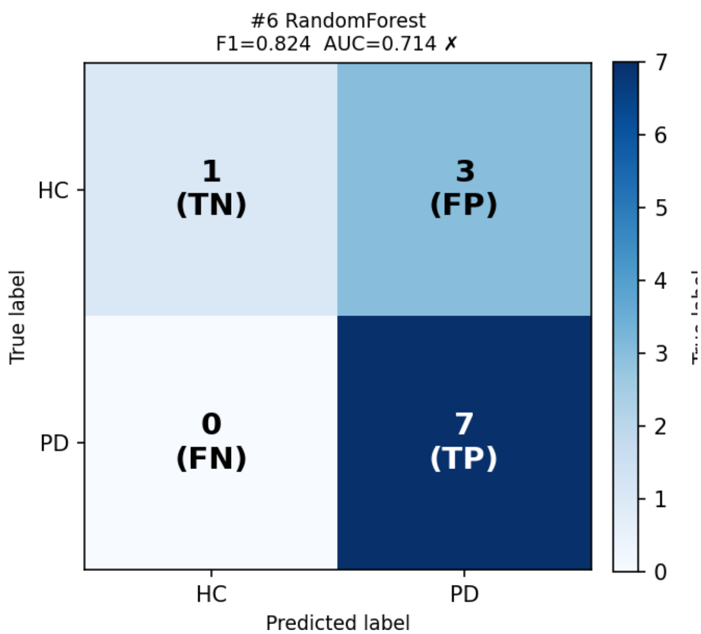


Figure 4-13 Random Forest Confusion Matrix

Classification report (Figure 4-14):

```
============================================================
 #6  RandomForest
 Test F1=0.8235  AUC=0.7143  Threshold=0.27
 Confusion: TN=1 FP=3 FN=0 TP=7
============================================================
              precision    recall  f1-score   support

          HC       1.00      0.25      0.40         4
          PD       0.70      1.00      0.82         7

    accuracy                           0.73        11
   macro avg       0.85      0.62      0.61        11
weighted avg       0.81      0.73      0.67        11
```

Figure 4-14 Random Forest Classification Report

Interpretation:

Ensemble learning in Random Forest provides some robustness to noise. However, due to the relatively small sample size, the diversity of individual decision trees is limited. The 40-feature selection generating the Random Forest helped focus on important connectivity measures.

➢ KNN

Configuration: The best-performing hyperparameters for the KNN model are:

- **n_neighbors:** 3
- **metric:** Manhattan
- **weights:** uniform
- **SelectKBest k:** 15
- **PCA variance:** 90%
- **Threshold:** 0.35

Test Set Performance:

KNN achieved F1=0.800 and AUC=0.804. Despite that KNN has the seventh best performance on the test set, KNN showed the best GridSearchCV score in F1 with a value of 0.832, implying good generalization skills while using cross-validation. The use of only 3 neighbors with

Manhattan distance implies high local decision boundaries, consistent with subject-level diversity in PD connectivity patterns. The sub-0.5 threshold (0.35) reflects the class imbalance and the KNN probability estimation method.

Confusion matrix (Figure 4-15):

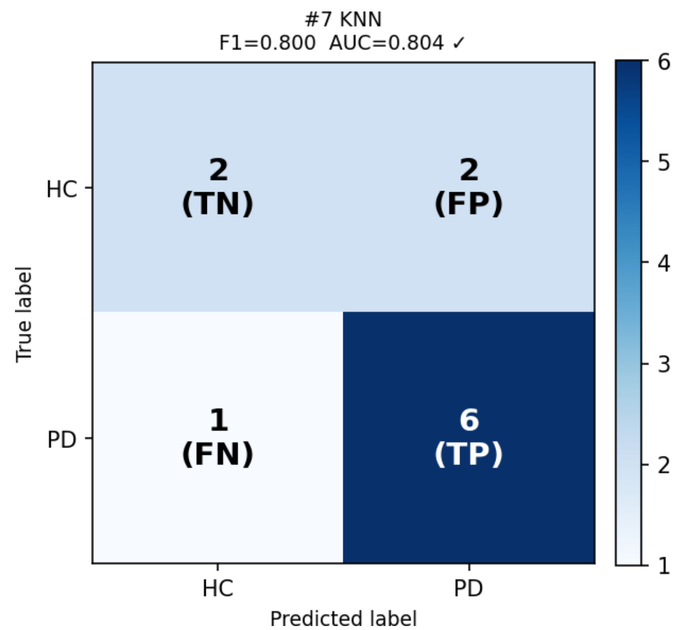


Figure 4-15 KNN Confusion Matrix

Classification report (Figure 4-16):

```
#7  KNN
Test F1=0.8000  AUC=0.8036  Threshold=0.35
Confusion: TN=2 FP=2 FN=1 TP=6

              precision    recall  f1-score   support

          HC       0.67      0.50      0.57         4
          PD       0.75      0.86      0.80         7

    accuracy                           0.73        11
   macro avg       0.71      0.68      0.69        11
weighted avg       0.72      0.73      0.72        11
```

Figure 4-16 KNN Classification Report

Interpretation:

The difference between the KNN CV F1 score (0.832) and the test F1 score (0.800) is the smallest among all models, indicating that KNN generalizes better than other classifiers. However, as shown in the repeated CV analysis (Section 4.3.3), the mean CV F1 of 0.678 ± 0.153 indicates high variance across different data splits.

➢ Extra Trees

Configuration The best-performing hyperparameters for the Extra Trees model are:

- **n_estimators:** 100
- **max_depth:** 4
- **min_samples_leaf:** 1
- **Class weight:** balanced
- **SelectKBest k:** 15
- **Threshold:** 0.39

Test Set Performance:

Extra Trees achieved F1=0.769 and AUC=0.786. It correctly identified the **most HC subjects (TN=3)** of any model, with only 1 false positive. However, this came with 2 false negatives (FN=2) the highest among the better-performing models. The compromise between HC and PD identification is clearly visible: Extra Trees favors HC specificity over PD sensitivity, an unusual pattern considering the class imbalance.

Confusion Matrix (Figure 4-17):

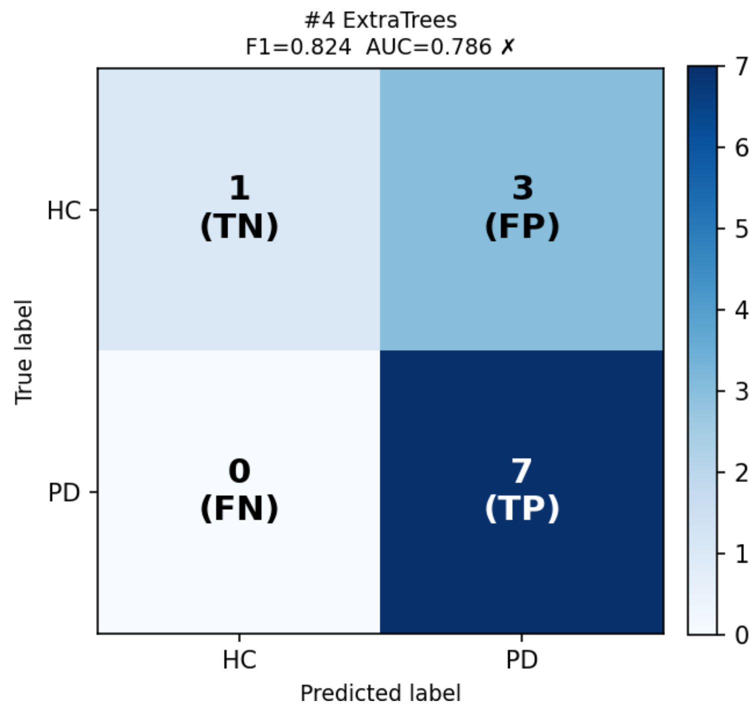


Figure 4-17 Extra Trees Confusion Matrix

Classification report (Figure 4-18):

```
=================================================================
 #4  ExtraTrees
 Test F1=0.8235  AUC=0.7857  Threshold=0.39
 Confusion: TN=1 FP=3 FN=0 TP=7
=================================================================
              precision    recall  f1-score   support

          HC       1.00      0.25      0.40         4
          PD       0.70      1.00      0.82         7

    accuracy                           0.73        11
   macro avg       0.85      0.62      0.61        11
weighted avg       0.81      0.73      0.67        11
```

Figure 4-18 Extra Trees Classification Report

Interpretation:

The strong class weight {HC:3, PD:1} effectively overcorrected the imbalance leading to excessive HC prediction. Extra Trees may benefit from further threshold calibration to balance sensitivity and specificity more evenly.

➢ XGBOOST

<u>Configuration:</u> The best performing hyperparameters for the XGBOOST model are:

- **n_estimators:** 200
- **max_depth:** 3
- **learning_rate:** 0.1
- **subsample:** 0.7
- **colsample_bytree:** 0.7
- **SelectKBest k:** 40
- **Threshold:** 0.45

<u>Test Set Performance:</u>

The XGBoost model scored an F1 of 0.750 and an AUC of 0.786, ranking ninth overall. Despite using scale_pos_weight to solve the class imbalance in a native way, XGBoost still produced TN=1 and FP=3, indicating that the native imbalance handling was insufficient. The low HC F1 of 0.33 (precision=0.50, recall=0.25) shows poor HC discrimination. XGBoost's regularized boosting approach, while powerful on large datasets, appears to be limited by the small training sample size of 44 subjects.

Confusion matrix (Figure 4-19):

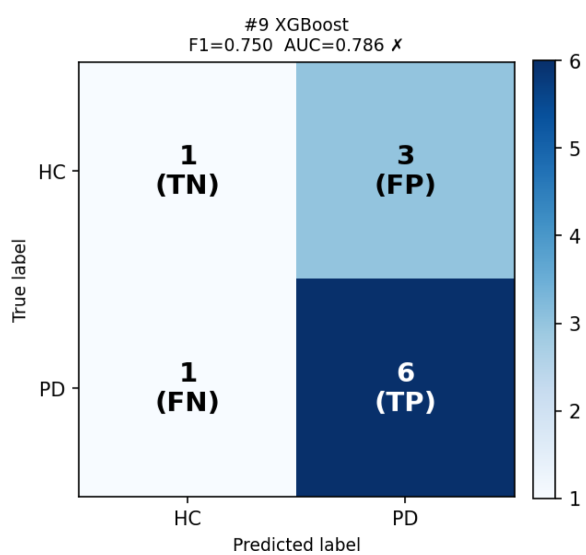


Figure 4-19 XGBoost Confusion Matrix

Classification report (Figure 4-20):

```
#9  XGBoost
Test F1=0.7500  AUC=0.7857  Threshold=0.45
Confusion: TN=1 FP=3 FN=1 TP=6
```

	precision	recall	f1-score	support
HC	0.50	0.25	0.33	4
PD	0.67	0.86	0.75	7
accuracy			0.64	11
macro avg	0.58	0.55	0.54	11
weighted avg	0.61	0.64	0.60	11

Figure 4-20 XGBoost Classification Report

Interpretation:

The XGBoost algorithm normally performs well with structured data with thousands of samples. The 55-subject dataset is inadequate for the XGBoost model, explaining the underperformance relative to simpler linear models.

➢ LightGBM

Configuration: The best performing hyperparameters for the LightGBM model are:

- **n_estimators:** 100
- **max_depth:** 3
- **learning_rate:** 0.05
- **num_leaves:** 15
- **Class weight:** balanced
- **SelectKBest k:** 15
- **Threshold:** 0.41

Test Set Performance:

LightGBM produced results identical to XGBoost on the test set (F1=0.750, AUC=0.786, TN=1), coming in the tenth place with the smallest GridSearchCV CV F1 of 0.675. Even with its superior computation time and leaf-wise splitting method, LightGBM did not outperform XGBoost. The identical confusion matrix to XGBoost suggests both models reached a similar decision boundaries on this dataset, both failing to distinguish HC subjects reliably.

Confusion matrix (Figure 4-21):

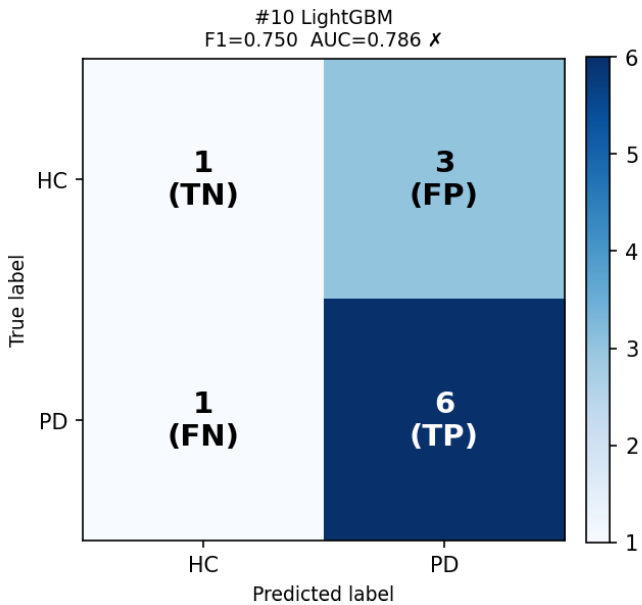


Figure 4-21 LightGBM Confusion Matrix

Classification report (Figure 4-22):

```
#10  LightGBM
Test F1=0.7500  AUC=0.7857  Threshold=0.41
Confusion: TN=1 FP=3 FN=1 TP=6

              precision    recall  f1-score   support

          HC       0.50      0.25      0.33         4
          PD       0.67      0.86      0.75         7

    accuracy                           0.64        11
   macro avg       0.58      0.55      0.54        11
weighted avg       0.61      0.64      0.60        11
```

Figure 4-22 LightGBM Classification Report

Interpretation:

LightGBM is efficient when dealing with large datasets with many features. The 55-subject dataset with 15–40 selected features there is not enough information for the leaf-wise splitting strategy to improve over simpler approaches. Both XGBoost and LightGBM are recommended only if the dataset is substantially expanded.

4.4 Repeated Stratified Cross-Validation — Top 3 Models + Soft Voting

To obtain stable, unbiased performance estimates, the top three individual models (SVM Linear, Logistic Regression, Ridge Classifier) and the Soft Voting ensemble were evaluated using 10 repeats × 5-fold stratified cross-validation (50 folds total). Each fold preserved the original HC/PD class ratio through stratified splitting.

Models were re-fitted using the training fold and evaluated on the held-out fold without any test-set leakage. SMOTE was applied only on the training folds, to avoid contamination of validation data.

The Soft Voting threshold (t = 0.35) was determined using 5-fold cross-validation on the full training set with the same threshold-search procedure used for the individual models, ensuring an unbiased comparison.

The Soft Voting ensemble combines the predicted probabilities from the three individual models (SVM Linear, Logistic Regression, and Ridge Classifier) by calculating their average. For each subject, the three predicted PD probabilities are summed and divided by three to produce a final average probability. If the final probability is greater than or equal to 0.35, the subject is classified as PD; otherwise, the subject is classified as HC.

4.4.1 Summary Results (mean ± std, 50 folds)

Table 4-2 presents the mean and standard deviation for F1-score, AUC, Precision, Recall, Accuracy, and mean True Negative (TN) count across all 50 folds. Green shading highlights the best value per metric.

Table 4-2 Repeated CV results — mean ± std over 50 folds (10 repeats × 5-fold). Green = best per metric. SoftVoting threshold = 0.35

Model	F1	AUC	Precision	Recall	Accuracy	TN (mean)
SVM_Linear	0.738 ± 0.123	0.716 ± 0.181	0.707 ± 0.116	0.788 ± 0.164	0.673 ± 0.141	2.200 ± 1.095
LogisticRegression	0.734 ± 0.124	0.736 ± 0.194	0.717 ± 0.125	0.769 ± 0.167	0.675 ± 0.139	2.340 ± 1.107
RidgeClassifier	0.727 ± 0.149	0.742 ± 0.183	0.729 ± 0.153	0.746 ± 0.189	0.676 ± 0.159	2.500 ± 1.204
SoftVoting	0.757 ± 0.123	0.732 ± 0.187	0.712 ± 0.123	0.825 ± 0.162	0.691 ± 0.138	2.140 ± 1.114

Interpretation:

The Soft Voting ensemble made the highest mean Recall (0.825 ± 0.162) and the highest F1-score (0.757 ± 0.123), indicating strong sensitivity toward PD detection across multiple data splits.

Ridge Classifier had the maximum mean value for the TN count (2.500 ± 1.204) and highest AUC (0.742 ± 0.183), indicating stronger protection against false positives and strong discriminative ability.

All models showed some standard deviation values (ranging between 0.12 and 0.19). This variation is expected due to the small sample size and small fold sizes (~11 subjects per fold), where even a single misclassified subject can noticeably shift the performance metrics by approximately 0.05–0.10 F1 units. This pattern is normal and expected in neuroimaging classification studies with limited sample sizes (n = 55) and has been reported in fMRI-based PD detection literature.

4.4.2 Detailed Numeric Summary

Table 4-3 provides the complete per-metric summary statistics (mean, standard deviation, minimum, maximum, and median) for all four models across the 50 evaluation folds.

Green shading highlights the two best metrics for clinical priority: Soft Voting Recall and Ridge Classifier TN count.

Table 4-3 Detailed numeric summary over 50 folds

Model	Metric	Mean	Std	Min	Max	Median
SVM_Linear	F1	0.738	0.123	0.400	1.000	0.750
	AUC	0.716	0.181	0.214	1.000	0.714
	Precision	0.707	0.116	0.429	1.000	0.714
	Recall	0.788	0.164	0.333	1.000	0.833
	Accuracy	0.673	0.141	0.364	1.000	0.682
	TN (mean)	2.200	1.095	0.000	4.000	2.000
LogisticRegression	F1	0.734	0.124	0.400	0.933	0.750
	AUC	0.736	0.194	0.107	1.000	0.758
	Precision	0.717	0.125	0.500	1.000	0.683
	Recall	0.769	0.167	0.333	1.000	0.833
	Accuracy	0.675	0.139	0.455	0.909	0.682
	TN (mean)	2.340	1.107	0.000	4.000	2.000
RidgeClassifier	F1	0.727	0.149	0.400	1.000	0.760

Model	Metric	Mean	Std	Min	Max	Median
RidgeClassifier	AUC	0.742	0.183	0.286	1.000	0.750
	Precision	0.729	0.153	0.429	1.000	0.714
	Recall	0.746	0.189	0.333	1.000	0.774
	Accuracy	0.676	0.159	0.364	1.000	0.727
	TN (mean)	2.500	1.204	0.000	5.000	3.000
SoftVoting	F1	0.757	0.123	0.400	1.000	0.774
	AUC	0.732	0.187	0.143	1.000	0.724
	Precision	0.712	0.123	0.500	1.000	0.700
	Recall	0.825	0.162	0.333	1.000	0.857
	Accuracy	0.691	0.138	0.455	1.000	0.727
	TN (mean)	2.140	1.114	0.000	4.000	2.000

Interpretation:

The minimum values indicate difficult folds in which hardcase subjects were put into validation, whereas the maximum values indicate that each model was capable to achieving perfect PD detection on favorable splits. Median values were generally higher than the means, meaning that some bad folds pulled down the average performance.

The figure below presents boxplots for F1, AUC, Precision, Recall, Accuracy, and TN across the four evaluated models (Figure 4-23).

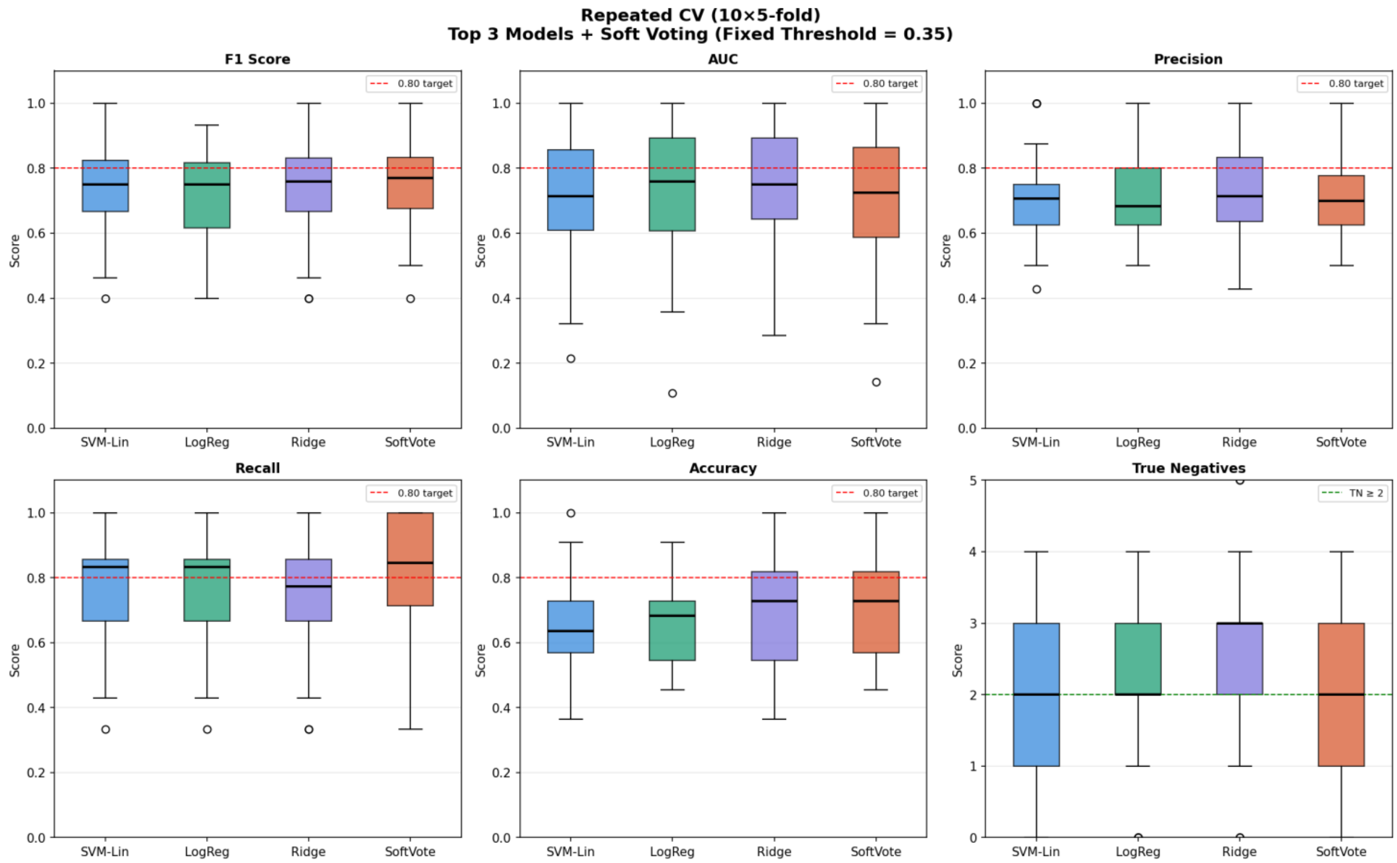


Figure 4-23 Boxplots for F1, AUC, Precision, Recall, Accuracy, and TN across the four evaluated models

4.4.3 Subject-Level Prediction Analysis — Top 3 Models + Soft Voting

Table 4-4 presents the prediction of each model for each of the 11 test subjects, evaluated on the final test set. Green cells indicate correct predictions (**v**) and red cells indicate incorrect predictions (**x**).

The bottom rows show the vote count for PD across the three individual models, the agreement percentage, the Soft Voting ensemble final prediction, and the average predicted probability P(PD).

Table 4-4 Subject-level predictions for the top three individual models and the Soft Voting ensemble.

Model	S01	S02	S03	S04	S05	S06	S07	S08	S09	S10	S11
Real Label	**HC**	**HC**	**PD**	**PD**	**PD**	**HC**	**PD**	**PD**	**PD**	**HC**	**PD**
SVM_Linear	PD ✗	HC ✓	PD ✓	PD ✓	PD ✓	HC ✓	PD ✓	PD ✓	PD ✓	PD ✗	PD ✓
LogisticRegression	HC ✓	HC ✓	PD ✓	PD ✓	PD ✓	HC ✓	PD ✓	PD ✓	PD ✓	PD ✗	HC ✗
RidgeClassifier	HC ✓	HC ✓	PD ✓	PD ✓	PD ✓	HC ✓	PD ✓	PD ✓	PD ✓	PD ✗	HC ✗
SoftVote	PD ✗	HC ✓	PD ✓	PD ✓	PD ✓	HC ✓	PD ✓	PD ✓	PD ✓	PD ✗	PD ✓
Votes for PD	**1**	**0**	**3**	**3**	**3**	**0**	**3**	**3**	**3**	**3**	**1**
Agreement %	33%	0%	100%	100%	100%	0%	100%	100%	100%	100%	33%
SoftVote Final	**PD**	**HC**	**PD**	**PD**	**PD**	**HC**	**PD**	**PD**	**PD**	**PD**	**PD**
Avg Probability	0.48	0.12	0.84	0.74	0.60	0.33	0.46	0.91	0.96	0.52	0.41

Interpretation:

There is an interesting clinical pattern to highlight. **Subjects S03, S04, S05, S07, S08, and S09** were correctly classified as PD by all models with average probabilities ranging from 0.46 to 0.96, and high agreement in all models, which implies that there are strong PD-related connectivity patterns. Subjects S01 and S10, both HC subjects, were misclassified as PD by the Soft Voting ensemble, indicating ambiguous connectivity patterns.

Misclassified HC subjects (S01, S10): Subject S01 (HC) was incorrectly predicted as PD by SVM Linear and the Soft Voting ensemble, with an average PD probability of 0.478, which is close to the decision threshold. However, Logistic Regression and Ridge Classifier correctly classified this subject as HC. Subject S10 (HC) was incorrectly classified as PD by all models, with an average PD probability of 0.524. This suggests that the brain connectivity pattern of this subject share similarities with PD subjects in the selected feature space. These two subjects can be considered difficult or borderline cases that may require further clinical evaluation.

Disagreement case (S11): Subject S11 (PD) was correctly classified by SVM Linear and the Soft Voting ensemble but incorrectly classified by Logistic Regression and Ridge Classifier. In this case, the Soft Voting ensemble was able to recover the correct prediction by combining the probabilities from all models, showing the advantage of the ensemble approach.

The figures below show the confusion matrix (Figure 4-25) and the classification report (Figure 4-24) of the Soft Voting ensemble model:

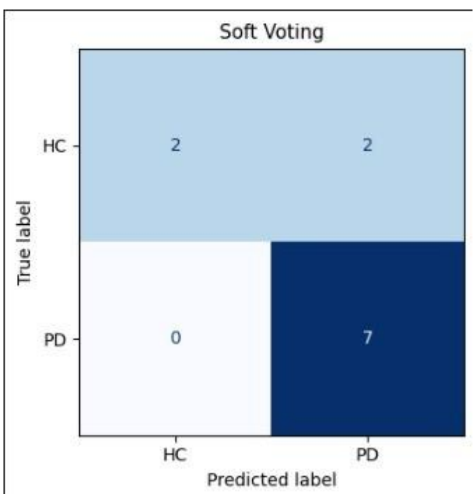
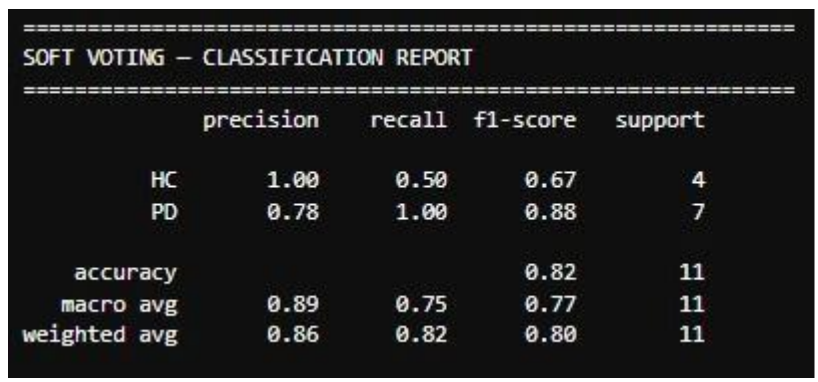


Figure 4-25 Soft Voting Confusion Matrix



SOFT VOTING – CLASSIFICATION REPORT

	precision	recall	f1-score	support
HC	1.00	0.50	0.67	4
PD	0.78	1.00	0.88	7
accuracy			0.82	11
macro avg	0.89	0.75	0.77	11
weighted avg	0.86	0.82	0.80	11

Figure 4-24 Soft Voting Classification Report

The figure below (Figure 4-25) shows the model agreement and P(PD) per subject, in the top 3 models and the Soft Voting ensemble model:

Figure 4-26 Model Agreement and P(PD) per subject, for the Top 3 models and the Soft Voting ensemble model

4.4.4 Per-Subject Summary

Table 4.16.2 summarizes the Soft Voting ensemble result for each subject individually, including the average ensemble probability and the vote count from the three individual models.

Table 4-5 Per-subject summary of Soft Voting ensemble predictions.

Subject	True Label	SoftVote	Avg Prob	Votes PD	Agree %	Result
S01	HC	PD x	0.478	1	33%	**WRONG**

Subject	True Label	SoftVote	Avg Prob	Votes PD	Agree %	Result
S02	HC	HC v	0.119	0	0%	**Correct**
S03	PD	PD v	0.839	3	100%	**Correct**
S04	PD	PD v	0.741	3	100%	**Correct**
S05	PD	PD v	0.595	3	100%	**Correct**
S06	HC	HC v	0.333	0	0%	**Correct**
S07	PD	PD v	0.465	3	100%	**Correct**
S08	PD	PD v	0.910	3	100%	**Correct**
S09	PD	PD v	0.956	3	100%	**Correct**
S10	HC	PD x	0.524	3	100%	**WRONG**
S11	PD	PD v	0.408	1	33%	**Correct**

Interpretation:

The Soft Voting ensemble correctly classified 9 out of 11 test subjects (Accuracy 81.8%), with 2 wrong predictions: both were HC subjects misclassified as PD (S01, S10). The two false positives are subjects with probabilities (0.478 and 0.524), both of which a clinician would likely state for further examination rather than treat as definitive PD diagnoses.

Importantly, all PD subjects were correctly identified (Recall= 1.0), resulting in zero false negatives on the held-out test set.

4.4.5 Clinical Interpretation and Final Model Selection

From a diagnostic perspective of Parkinson's Disease, Recall (Sensitivity) is considered the most clinically important metric because missing a true PD patient may delay diagnosis and can result in late detection and treatment. For this reason, the Soft Voting ensemble proved to be the strongest in terms of clinical suitability by achieving the highest Recall (0.825 during repeated cross-validation and 1.00 on the held-out test set).

Specificity and True Negative count continue to be important secondary metrics because they reduce unnecessary tests on healthy subjects. Ridge Classifier achieved the highest mean TN count (2.500) and the highest AUC (0.742), indicating stronger separation between HC and PD subjects.

The Soft Voting ensemble model also had the highest F1-score (0.757), showing balanced classification performance. Furthermore, the subject-level analysis indicated that the ensemble successfully corrected difficult disagreement cases through probability averaging.

Based on the combined repeated cross-validation analysis, subject-level analysis, and clinical metric interpretation, the Soft Voting ensemble is selected as the primary recommended model for Parkinson's disease prediction, while Ridge Classifier is considered the strongest secondary model due to its high AUC and healthy-subject protection performance.

Overall, the result of this project indicates that resting-state MRI functional connectivity data can be effectively used for Parkinson's disease prediction using machine learning techniques. The obtained performance, especially the high Recall and stable cross-validation results, proves that machine learning models can provide reliable support for early Parkinson's disease detection.

5 Perspectives

This project proves the feasibility of using resting-state fMRI scans data and machine learning techniques for Parkinson's disease prediction. However, several improvements can be explored in future work.

One important improvement would be enlarging the size of the dataset used, since increasing the number of subjects could improve model generalization, reduce performance variance, and allow the models to learn more representative brain connectivity patterns.

Future studies may also investigate more advanced neuroimaging preprocessing techniques that could improve the quality of the extracted data and potentially lead to better modeling performance.

In addition, applying more advanced feature engineering and feature extraction techniques can help generate more informative connectivity features and improve classification accuracy. Exploring deep learning approaches and additional neuroimaging modalities could also further enhance the prediction performance and clinical reliability of the system.

6 References

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